

Cubic Polynomial Trajectory Planning

1 Cubic trajectory

Consider a cubic polynomial trajectory defined over the time interval $t \in [t_s, t_f]$:

$$q(t) = a_3 t^3 + a_2 t^2 + a_1 t + a_0. \quad (1)$$

Its velocity and acceleration are

$$\dot{q}(t) = 3a_3 t^2 + 2a_2 t + a_1, \quad (2)$$

$$\ddot{q}(t) = 6a_3 t + 2a_2. \quad (3)$$

The four polynomial coefficients are determined by prescribing the initial and final positions and velocities:

$$q(t_s) = q_s, \quad \dot{q}(t_s) = \dot{q}_s, \quad (4)$$

$$q(t_f) = q_f, \quad \dot{q}(t_f) = \dot{q}_f. \quad (5)$$

Substitution into the cubic polynomial gives the four boundary-condition equations

$$a_3 t_s^3 + a_2 t_s^2 + a_1 t_s + a_0 = q_s, \quad (6)$$

$$a_3 t_f^3 + a_2 t_f^2 + a_1 t_f + a_0 = q_f, \quad (7)$$

$$3a_3 t_s^2 + 2a_2 t_s + a_1 = \dot{q}_s, \quad (8)$$

$$3a_3 t_f^2 + 2a_2 t_f + a_1 = \dot{q}_f. \quad (9)$$

In matrix form, these equations become

$$\underbrace{\begin{bmatrix} t_s^3 & t_s^2 & t_s & 1 \\ t_f^3 & t_f^2 & t_f & 1 \\ 3t_s^2 & 2t_s & 1 & 0 \\ 3t_f^2 & 2t_f & 1 & 0 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} a_3 \\ a_2 \\ a_1 \\ a_0 \end{bmatrix}}_{\mathbf{a}} = \underbrace{\begin{bmatrix} q_s \\ q_f \\ \dot{q}_s \\ \dot{q}_f \end{bmatrix}}_{\mathbf{b}}. \quad (10)$$

Therefore, the polynomial coefficients can be computed as

$$\boxed{\mathbf{a} = \mathbf{A}^{-1} \mathbf{b}}. \quad (11)$$

In numerical implementations, the linear system $\mathbf{A}\mathbf{a} = \mathbf{b}$ should normally be solved directly instead of explicitly forming $\mathbf{A}^{-1}$.

2 Solution using shifted time

The closed-form solution is simpler if time is measured from the beginning of the motion. Define

$$\tau = t - t_s, \quad T = t_f - t_s, \quad (12)$$

so that $\tau \in [0, T]$. Write the trajectory as

$$q(\tau) = a_0 + a_1\tau + a_2\tau^2 + a_3\tau^3. \quad (13)$$

The velocity and acceleration are

$$\dot{q}(\tau) = a_1 + 2a_2\tau + 3a_3\tau^2, \quad (14)$$

$$\ddot{q}(\tau) = 2a_2 + 6a_3\tau. \quad (15)$$

The boundary conditions now read

$$q(0) = q_s, \quad \dot{q}(0) = \dot{q}_s, \quad (16)$$

$$q(T) = q_f, \quad \dot{q}(T) = \dot{q}_f. \quad (17)$$

After substitution, one obtains

$$a_0 = q_s, \quad (18)$$

$$a_1 = \dot{q}_s, \quad (19)$$

$$a_0 + a_1T + a_2T^2 + a_3T^3 = q_f, \quad (20)$$

$$a_1 + 2a_2T + 3a_3T^2 = \dot{q}_f. \quad (21)$$

Equivalently,

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & T & T^2 & T^3 \\ 0 & 1 & 2T & 3T^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} q_s \\ \dot{q}_s \\ q_f \\ \dot{q}_f \end{bmatrix}. \quad (22)$$

Solving this system gives

$$a_0 = q_s, \quad (23)$$

$$a_1 = \dot{q}_s, \quad (24)$$

$$a_2 = \frac{3(q_f - q_s)}{T^2} - \frac{2\dot{q}_s + \dot{q}_f}{T}, \quad (25)$$

$$a_3 = -\frac{2(q_f - q_s)}{T^3} + \frac{\dot{q}_s + \dot{q}_f}{T^2}. \quad (26)$$

Consequently, the complete trajectory is

$$\boxed{\begin{aligned} q(t) &= q_s + \dot{q}_s(t - t_s) \\ &+ \left[\frac{3(q_f - q_s)}{T^2} - \frac{2\dot{q}_s + \dot{q}_f}{T} \right] (t - t_s)^2 \\ &+ \left[-\frac{2(q_f - q_s)}{T^3} + \frac{\dot{q}_s + \dot{q}_f}{T^2} \right] (t - t_s)^3, \end{aligned}} \quad T = t_f - t_s. \quad (27)$$

3 Zero initial and final velocities

If the trajectory must start and finish at rest,

$$\dot{q}_s = \dot{q}_f = 0, \quad (28)$$

the coefficients reduce to

$$a_0 = q_s, \quad a_1 = 0, \quad a_2 = \frac{3(q_f - q_s)}{T^2}, \quad a_3 = -\frac{2(q_f - q_s)}{T^3}. \quad (29)$$

The corresponding trajectory is

$$\boxed{q(t) = q_s + 3(q_f - q_s) \left(\frac{t - t_s}{T} \right)^2 - 2(q_f - q_s) \left(\frac{t - t_s}{T} \right)^3}. \quad (30)$$

Introducing the normalized time

$$s = \frac{t - t_s}{T}, \quad s \in [0, 1], \quad (31)$$

the same result can be written compactly as

$$\boxed{q(s) = q_s + (q_f - q_s)(3s^2 - 2s^3)}. \quad (32)$$